

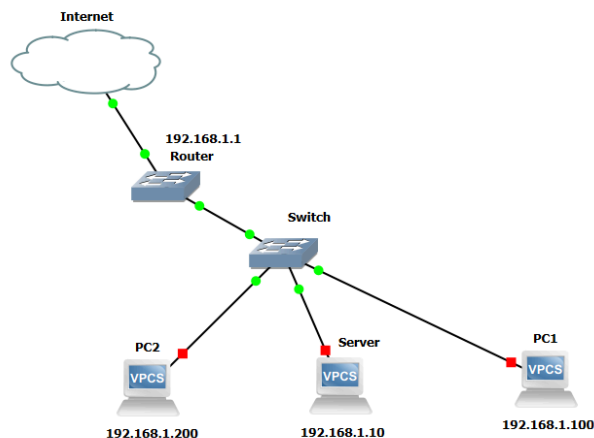
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CSCE 3530
Computer Network Design Project Proposal

Overview

The goal is to create a working client-server network where devices can communicate using the core networking protocols, and then demonstrate and analyze that traffic using Wireshark.

Network Setup/Design

My planned topology is a single internal LAN connected to a simulated external network. It includes router, switch, client PCs and server. The router sits at the edge of the network and acts as the gateway. It will handle DHCP, NAT and routing between the internal network and the outside. The switch connects all the internal devices together. The two client PCs will receive their IP addresses automatically through DHCP, and the server will run a static address so the client can reach it for web and DNS services. A cloud represents the internet connection so I can demonstrate traffic moving between the private network and an external network.



The topology shown in the figure is a baseline design. I plan to expand it by adding more client PCs along with additional servers to the internal LAN to better demonstrate different protocols.

Basic Networking Services

- 1) DHCP: Router will demonstrate this protocol by handing out IP addresses to the PCs automatically.
- 2) DNS: Configures user-friendly domain names to the specific static IP
- 3) ARP: Devices discovers each other's MAC addresses

- 4) HTTP/HTTPS: Server will run a web server for HTTP/HTTPS communication
- 5) SSL/TLS: Demonstrated through HTTPS to show the encrypted version of web traffic
- 6) NAT: Translating private addresses to a public one
- 7) ICMP: Using ping and trace route to test connectivity
- 8) TCP & UDP: Captured in Wireshark to compare connect based and connectionless traffic
- 9) IP addressing and subnetting: The internal LAN will be designed using a structured private addressing scheme to ensure logical boundaries and efficient address allocation.

Tools and Testing

I am using GNS3 as the main simulator, along with VirtualBox to run the GNS3 virtual machine and any guest systems. For the router I am using a free, openly available router image. I will use Wireshark to capture and analyze packets, and standard tools like ping and traceroute to verify connectivity. My testing plan is to first confirm every device can reach the others, then bring up each service (DHCP, DNS, HTTP/HTTPS) and confirm it works and finally capture the traffic in Wireshark to observe how DHCP, DNS, ARP, TCP, and UDP actually behave on the wire.